FROM SCRATCH LEARNING: BREAST IMAGES ON A IMBALANCED TO TRANSFER CLASSIFYING ULTRASOUND SMALL DATASET



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*click the figure for project repository

FDL Project, A.A. 2025/2026

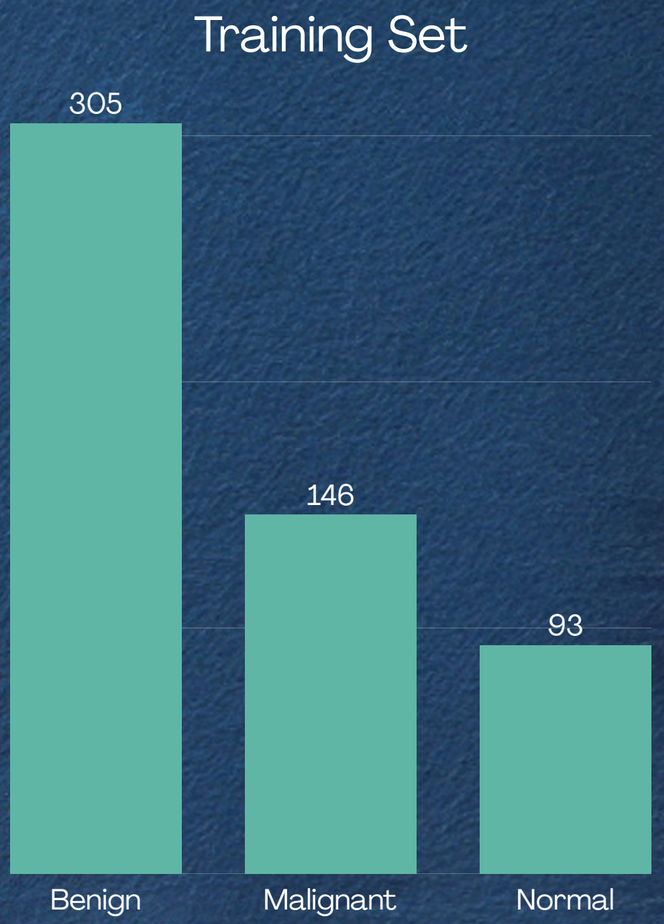
PROJECT GOAL

CNNs from Scratch vs Transfer Learning

- Task: Breast ultrasound image **classification**
- Dataset: BUSI (600 female patients, ages 25-75)
- 3 classes:
  - **normal** (no findings)
  - **benign** (non-cancerous mass),
  - **malignant** (cancerous mass)

CAN A CNN TRAINED FROM SCRATCH LEARN ROBUST FEATURES FROM A SMALL MEDICAL DATASET, OR IS TRANSFER LEARNING MORE EFFECTIVE?

DATA LOADING AND CLASS IMBALANCE



- Original dataset size: 780 images (masked images removed)
- Split size: 70% - 15% - 15% {classwise to preserve target class distribution}
- Images resized to 224 × 224 pixels during loading

MAIN ISSUES:

- small
- imbalanced

{CLASS WEIGHTS}

- **WHY** ? Training set imbalanced
- Class weights **modify the loss**, but they do not create new images

$$w_c = \frac{N}{K \cdot n_c}$$

- **N: number of training samples**
- **K: number of classes (3)**
- **Nc: number of samples in class c**

CLASS	WEIGHT
Benign	**0.5945**
Malignant	**1.2420**
Normal	**1.9498**

- Weights will be passed in model.fit(class_weight=class_weight_dict)

{AUGMENTATION & DUPLICATION}

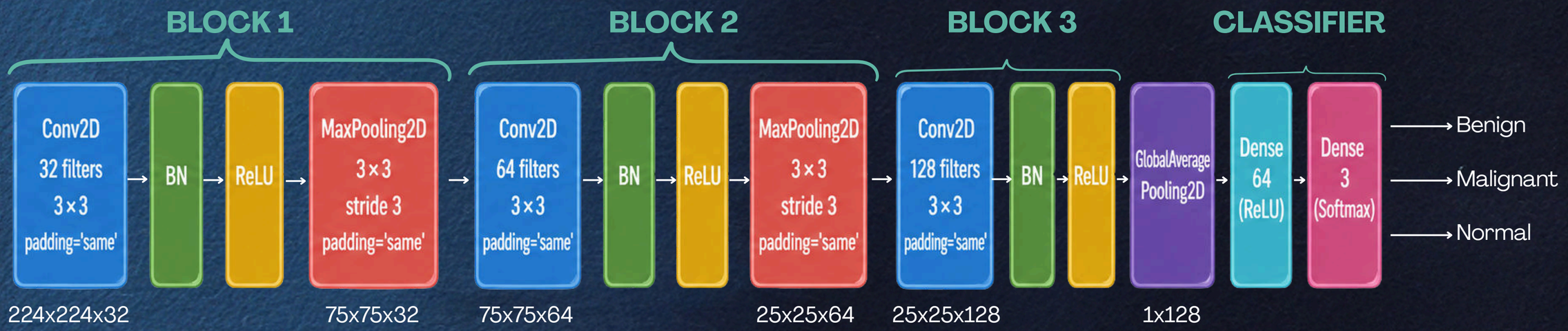
- **Augmentation***
  - **Rotation**: simulates minor probe **orientation variations**
  - **Brightness**: simulates **exposure variations** between acquisitions
  - **Contrast**: simulates **gain variations** between ultrasound machines
  - **Horizontal flip***: anatomically acceptable (**left/right breast symmetry**)
- **Duplication***
  - benign: x1
  - malignant: x2
  - normal: x3

1* **Inline aug.** is applied during training at each epoch, **variability grows, but train size fixed** (**544**); **offline aug.** physically generates and saves extra copies (**3 x image + original**), **train size up to 2176**. **Both versions still produce imbalanced dataset.**

2* Vertical flip was not applied because it could distort the natural orientation of the ultrasound images.

3* Training set size **after duplication** is **876**.

CNN FROM SCRATCH ARCHITECTURE



Training Setup

- Optimizer: RMSprop
- Learning rate: 0.001
- EarlyStopping: patience = 10

WHY GLOBAL AVERAGE POOLING?

- Uses the whole feature map
- Distributes gradients across every spatial position

Global Max Pooling, **instead**:

- Keeps only the maximum activation
- Gradients flow only through max locations

WHY ? On a small dataset, **GAP helps exploit more learning signal from each image**

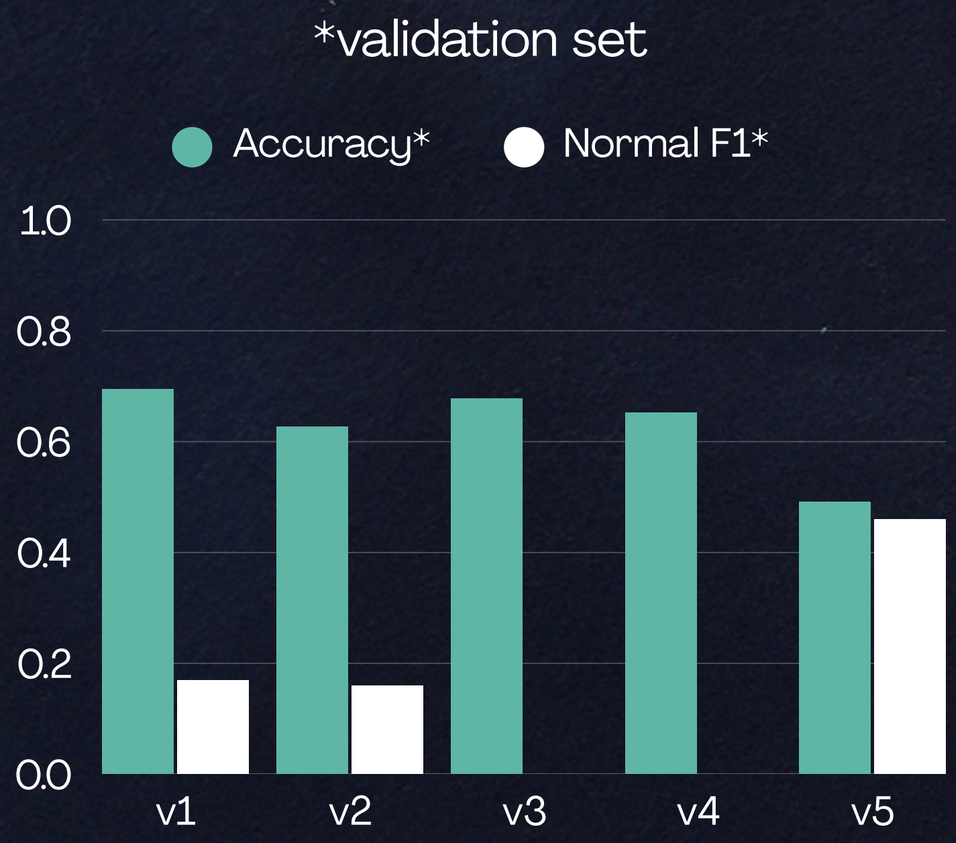
REBALANCING STRATEGIES

Version of cnn from scratch	TrainingSet Strategy	Rotation	Contrast & Brighness	Dropout (GAP & Dense)	Class weight
V1	Original Dataset (Inline Augumentation)	5%	20%	None	✓
V2	Original Dataset (Inline Augumentation)	8%	30%	30%	✓
V3	Oversampling (identical copies x2/x3)	8%	30%	30%	✗
V4	Offline Augumentation (3 aug copies / img)	8%	30%	50%	✓
V5	Offline Augumentation (3 aug copies / img)	8%	30%	None	✓

NORMAL CLASS DETECTION COLLAPSE

Why? Mitigations treat the symptom, not the cause. Augmentation, class weights, oversampling and dropout help only partially. None of these compensate for learning low-level features

The **root problem remains**: too few images to learn robust visual features from scratch.



TRANSFER LEARNING APPROACH

Pre-trained CNNs already learned general visual features:

- Edges
- Textures
- Local patterns
- Shapes

ImageNet can still help? Even if It is not medical, early visual features are general and reusable.

⟶ **reducing trainable parameters and limiting overfitting risk**

FOUR PRETRAINED BACKBONES

Frozen ImageNet weights + shared head: Dense(128, ReLU) → Dropout(0.3) → Softmax(3)

VGG16

Parameters
138M

EfficientNetB0

Parameters
5.3M

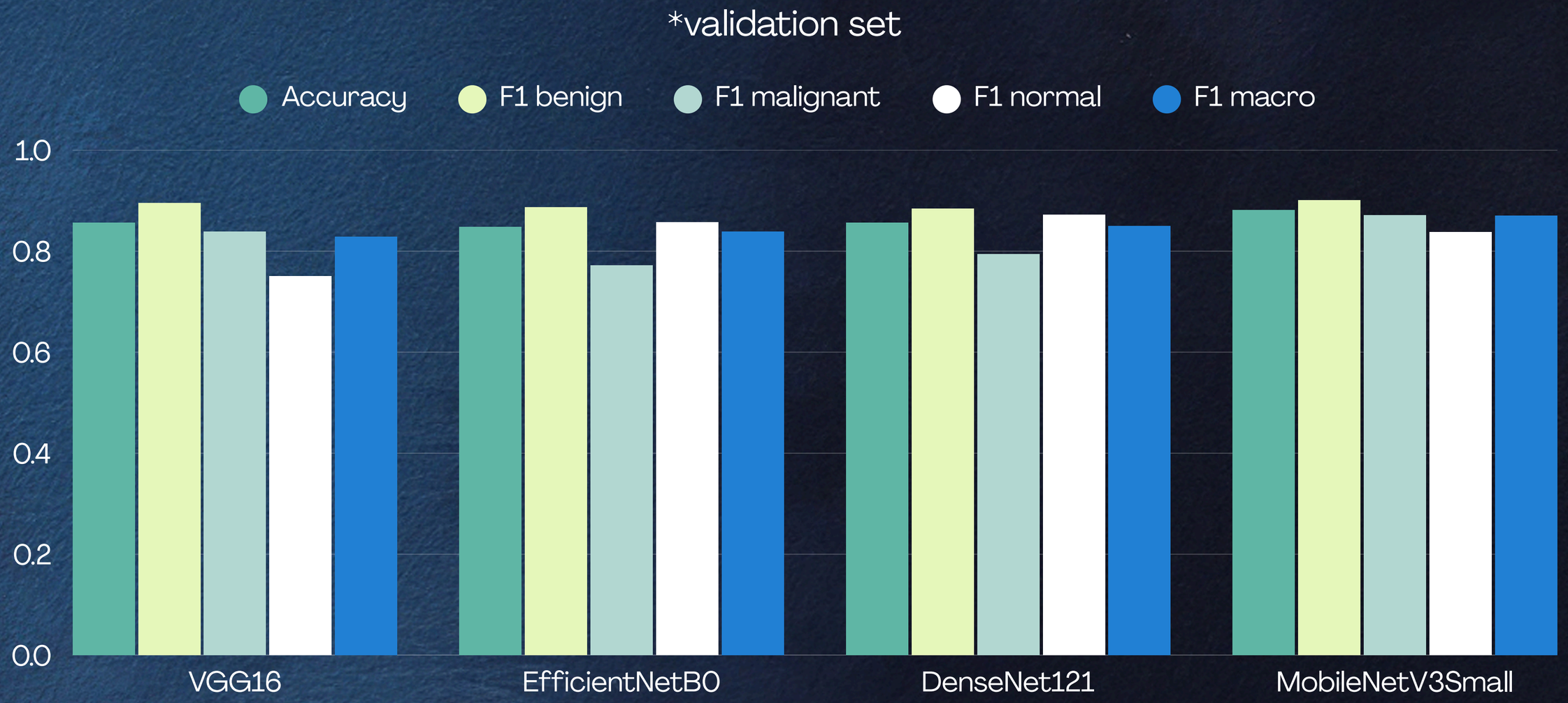
DenseNet121

Parameters
8M

MobileNetV3Small

Parameters
2.5M

TRANSFER LEARNING RESULTS*



V1 MODEL VS MOBILENETV3SMALL*

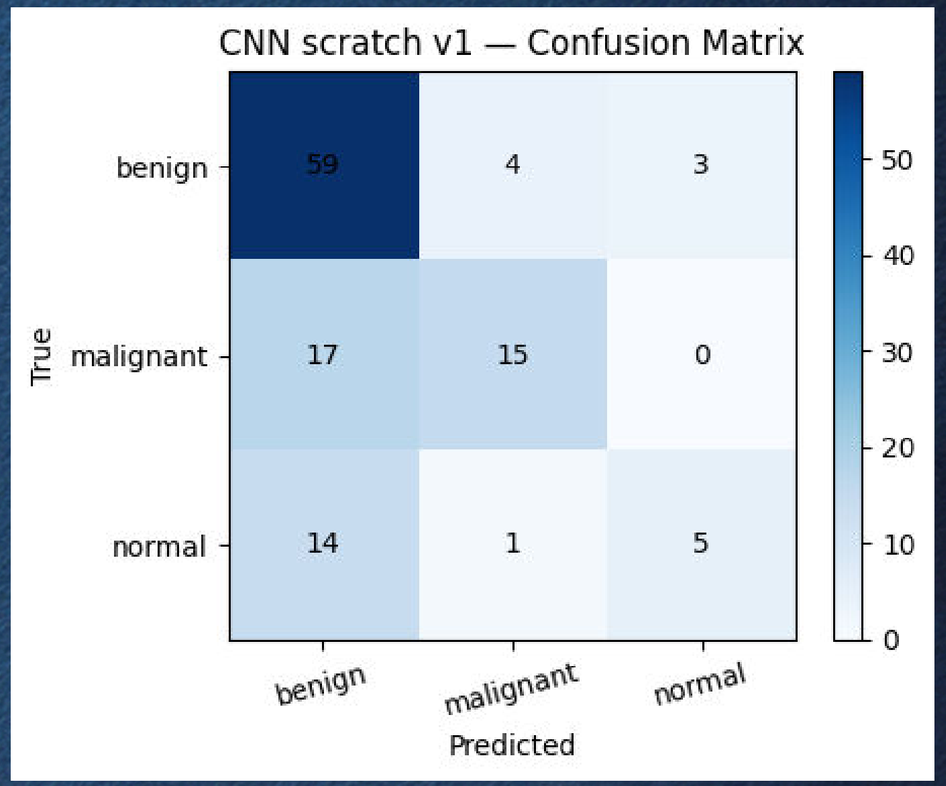
*TESTSET

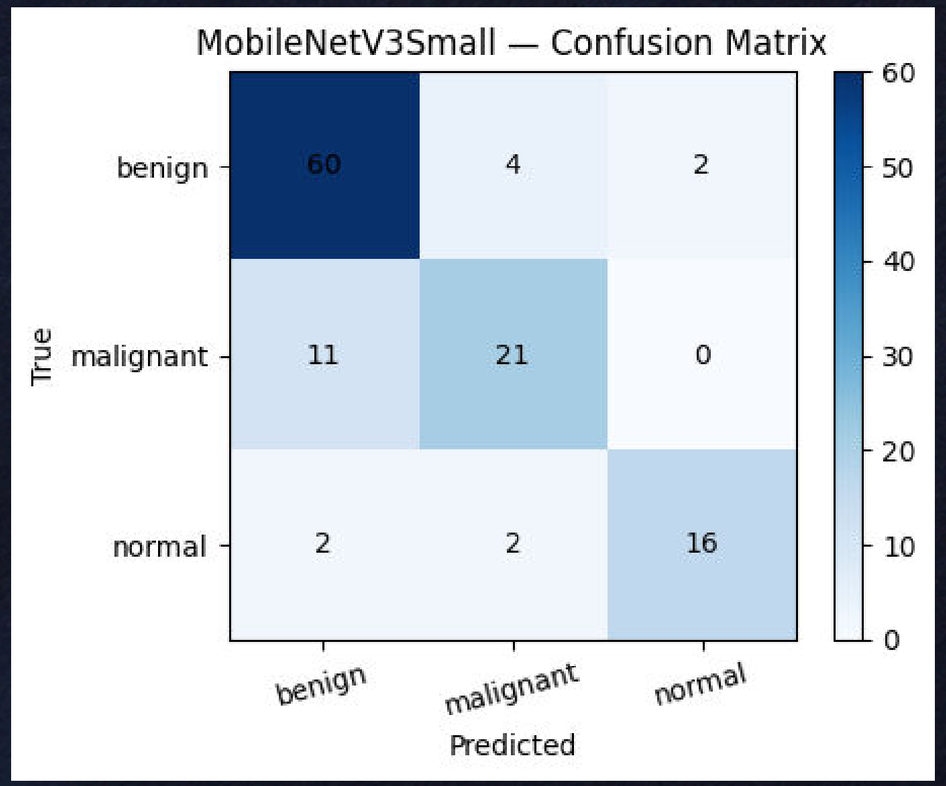
MODEL	ACCURACY	F1 MACRO
Scratch best (V1)	0.67	0.56
MobileNetV3	0.82	0.81

- Malignant → benign confusion persists in both approaches

V1 MODEL VS MOBILENETV3SMALL*

*TESTSET





MAIN CONCLUSION AND LIMITATIONS

- Not acceptable performance levels for a clinical support tool
- A from-scratch CNN on 544 images tops out around 0.67 accuracy
- Transfer learning with MobileNetV3Small raises test accuracy to 0.82 and essentially solves the normal class, without any new data
- Malignant → benign confusion persists in both approaches

- **The binding constraint is data scarcity (well known bottleneck in medical imaging)**